Validation

Hostel Management System (HMS) — Software Requirements Specification

Purpose

Define shared input-validation rules for data entered or imported into the Hostel Management System. Validation protects data quality and supports consistent processing across student, hostel, room, bed, application, allotment, renewal, complaint, payment, visitor, and notice workflows.

Validation principles

- Validate all untrusted input on the server, even when client-side validation is also provided.
- Use field-level rules and business rules. Normalize values consistently before validation, while preserving the original value when it is required for display or audit purposes.
- Reject invalid data with clear field-specific feedback. Do not silently accept or truncate values in a way that changes their meaning.
- Use parameterized database operations and context-appropriate output encoding; validation alone is not a substitute for injection protection.

Common field rules

- Required fields must not be empty or contain only whitespace.
- Apply documented minimum and maximum lengths to names, identifiers, addresses, descriptions, and comments.
- Validate email addresses and contact numbers using reasonable format rules; do not assume one country's phone format unless the project requires it.
- Validate dates using an unambiguous format and reject impossible dates. Check date relationships, such as application, check-in, checkout, and payment dates, against the relevant workflow rules.
- Validate numeric values as numbers and enforce allowed ranges, precision, and non-negative constraints where appropriate.

HMS-specific business rules

- Student identifiers and account identifiers must be unique where the data model requires uniqueness.
- A room's capacity must be a positive value. A bed can have only one active occupant/allotment at a time.
- An allotment must reference an existing eligible student and an available bed/room in the relevant hostel.
- Application review, renewal, transfer, check-in, and check-out actions must follow allowed status transitions.
- Payment amount and payment status must satisfy the configured fee workflow; do not treat a client-submitted payment status as proof that payment was received.
- Visitor records must include the required host/student reference and visit details defined by the workflow.

Cross-field and state validation

- Check relationships between fields, not only individual formats. For example, checkout must not precede check-in, and an end date must not precede a start date.
- Re-check state-dependent rules immediately before saving to avoid race conditions, such as two requests attempting to allocate the same bed.

- Use database constraints and transactions to enforce critical uniqueness and consistency rules in addition to application validation.

Error reporting

- Return a clear message identifying the field or business rule that failed.
- Avoid revealing internal database structure or sensitive record information in validation messages.
- Preserve safe, valid form values after validation failure where practical.

Acceptance criteria

- Invalid and missing required values are rejected on the server.
- Uniqueness, capacity, availability, and workflow-state rules are enforced consistently.
- Concurrent requests cannot create conflicting active bed allocations.
- Validation feedback is understandable and does not disclose internal system details.

Document note: This document defines project-level requirements for the HMS utility. Confirm implementation-specific settings (such as role names, file limits, retention periods, and timeout values) against the approved project configuration before production deployment.